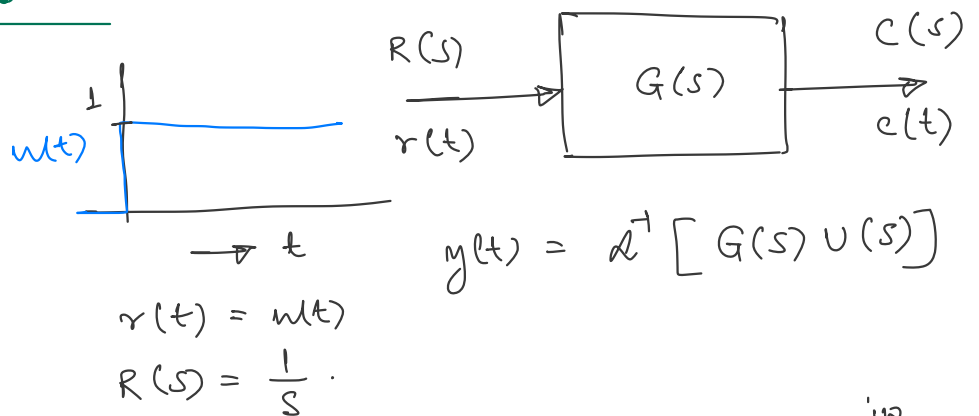


Time response of LTI systems

First order systems

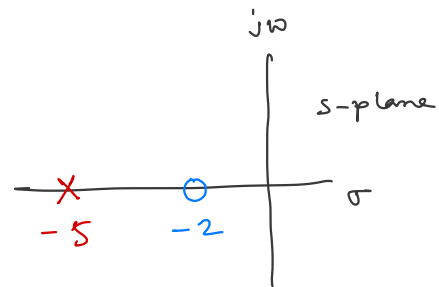


$$y(t) = \mathcal{L}^{-1} [G(s) U(s)]$$

$$G(s) = \frac{s+2}{s+5}$$

$$\text{pole: } -5$$

$$\text{zero: } -2$$



$$C(s) = \frac{s+2}{s(s+5)} = \frac{A}{s} + \frac{B}{s+5} =$$

$$A = \left. \frac{s+2}{s+5} \right|_{s=0} = \frac{2}{5} ; \quad B = \left. \frac{s+2}{s} \right|_{s=-5} = \frac{3}{5}$$

$$c(t) = \underbrace{\frac{2}{5} u(t)}_{\text{forced response}} + \underbrace{\frac{3}{5} e^{-5t} u(t)}_{\text{natural response}}$$

Observations:

- The root of the Laplace transformed input function's denominator decides the form of the forced response.

- The root of the system's transfer function's denominator ($s = -5$) generates the form of the natural response

— $e^{-\alpha t}$, where $-\alpha$ is the pole

— if $(-\alpha)$ is more negative, decay exponentially even faster.

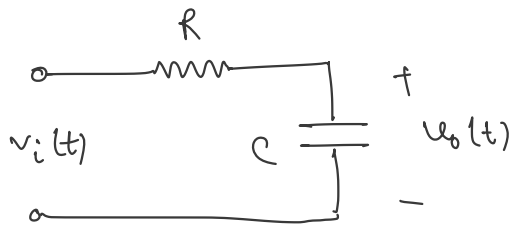
- o The roots of both the numerator and denominator generate the amplitude of the responses.

$$R(s)G(s) = \frac{N(s)}{s(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3} + \frac{D}{s}$$

$$R(s) = \frac{1}{s}$$

$$c(t) = D u(t) + A e^{-t} u(t) + B e^{-2t} u(t) + C e^{-3t} u(t).$$

Example.



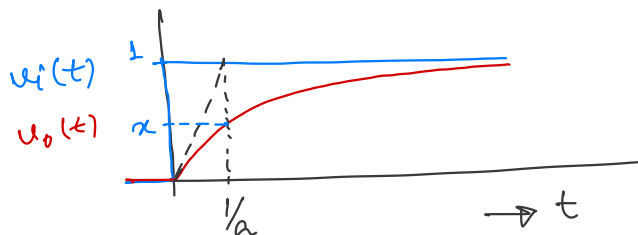
$$\frac{v_o(s)}{v_i(s)} = \frac{1/sC}{R + 1/sC} = \frac{1/RC}{s + 1/RC} = \frac{a}{s + a}$$

$$a := \frac{1}{RC}$$

$$v_i(t) = u(t) ; \quad v_i(s) = \frac{1}{s}$$

$$G(s) = \frac{a}{s + a} ; \quad c(s) = \frac{a}{s(s + a)}$$

$$c(t) = (1 - e^{-at}) u(t)$$



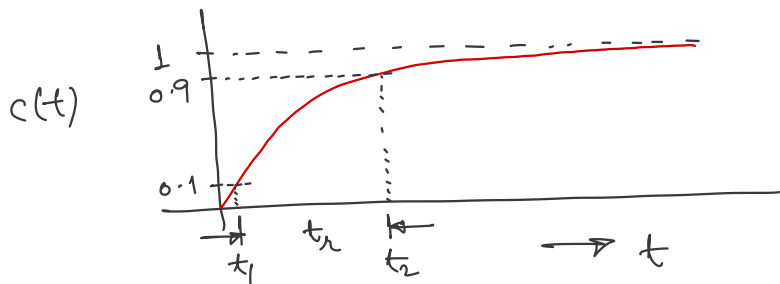
Initial slope of the response $c(t)$: $\left. \frac{dc(t)}{dt} \right|_{t=0^+} = a$.

$$c(t = \frac{1}{a}) = 1 - e^{-1} \approx 0.632$$

* Time constant: time the system takes for the step response to rise to 63% of its final value (here 1)

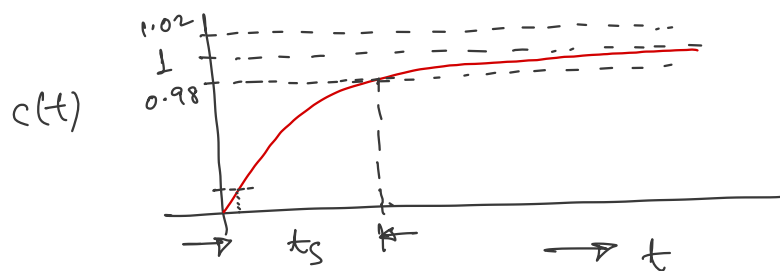
$$\text{time-constant} = \frac{1}{|a|} \equiv \frac{1}{|\text{pole location}|}$$

* Rise time: time for the response to ^{rise} from 10% to 90% of its final value.



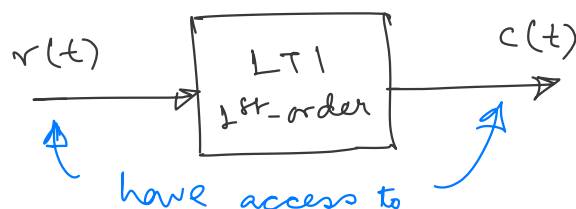
$$\left. \begin{aligned} c(t_1) &= 1 - e^{-at_1} = 0.1 \\ c(t_2) &= 1 - e^{-at_2} = 0.9 \end{aligned} \right\} t_r = t_2 - t_1 \approx \frac{2.2}{a}$$

* Settling time: time for the response to reach, and stay within 2% of its final value (2% settling time)



$$c(t) \Big|_{t=t_s} = 0.98 = 1 - e^{-at_s} \Rightarrow T_s = \frac{4}{a}$$

* First order system identification



Assume $G(s) = \frac{k}{s+a}$

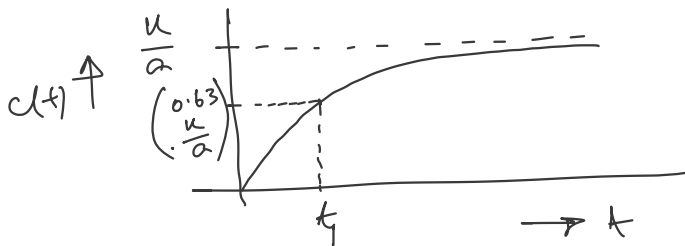
Apply $r(t) = u(t)$;

$$C(s) = \frac{k}{s(s+a)}$$

$$= \frac{k}{a} \cdot \frac{1}{s} - \frac{k/a}{s+a}$$

$$c(t) = \frac{k}{a} [1 - e^{-at}] u(t)$$

final value $= \frac{k}{a}$, time constant $= \frac{1}{a} = \tau$



● Second order systems:

Let $G(s) = \frac{b}{s^2 + as + b}$

Poles: $\frac{-a \pm \sqrt{a^2 - 4b}}{2} = -\frac{a}{2} \pm \sqrt{\left(\frac{a}{2}\right)^2 - b}$

(assume $b > 0$) (also $a \geq 0$).

always true for "stable" systems

$$= -\underbrace{\frac{a}{2\sqrt{b}}}_{\xi} \cdot \underbrace{\sqrt{b}}_{\omega_n} \pm \sqrt{b} \cdot \sqrt{\left(\frac{a}{2\sqrt{b}}\right)^2 - 1}$$

$$= -\xi \omega_n \pm \omega_n \sqrt{\xi^2 - 1}$$

$\xi = \frac{a}{2\sqrt{b}}$ $\therefore$ damping ratio

$\omega_n = \sqrt{b}$ $\therefore$ (undamped) natural frequency

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

← Standard 2nd order t.f.

Case-I : Real poles : $\frac{\alpha}{2\sqrt{b}} \geq 1 \Rightarrow \zeta \geq 1$.

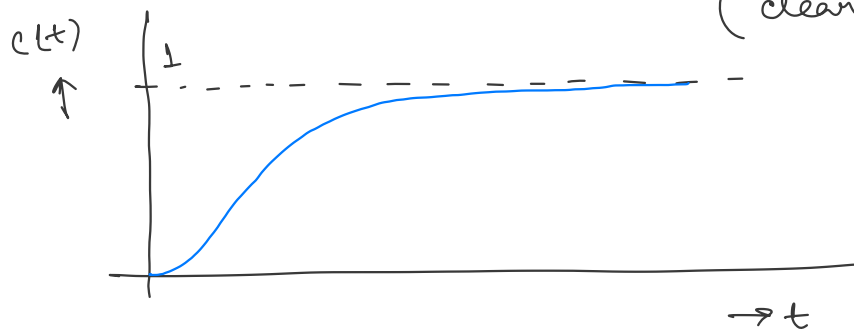
case - Ia : Distinct real poles ($\zeta > 1$)

$$\begin{aligned} \text{let } \sigma_1 &= -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1} \\ \sigma_2 &= -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} \end{aligned}$$

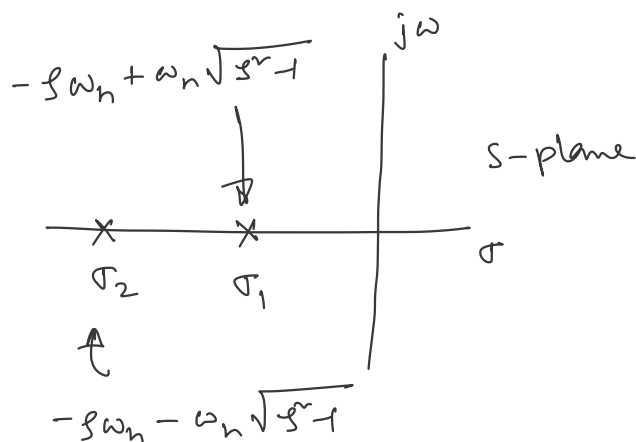
$$\begin{aligned} R(s) = \frac{1}{s} &\Rightarrow C(s) = \frac{\omega_n^2}{s(s - \sigma_1)(s - \sigma_2)} \\ &= \frac{k_1}{s} + \frac{k_2}{s - \sigma_1} + \frac{k_3}{s - \sigma_2} \end{aligned}$$

$$c(t) = \left[k_1 + k_2 e^{\sigma_1 t} + k_3 e^{\sigma_2 t} \right] u(t)$$

(clearly $k_1 = 1$)



overdamped



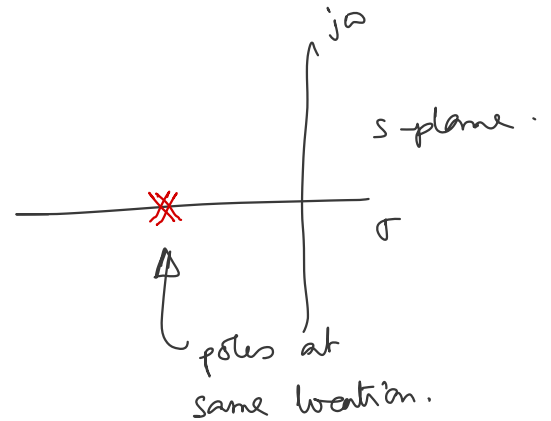
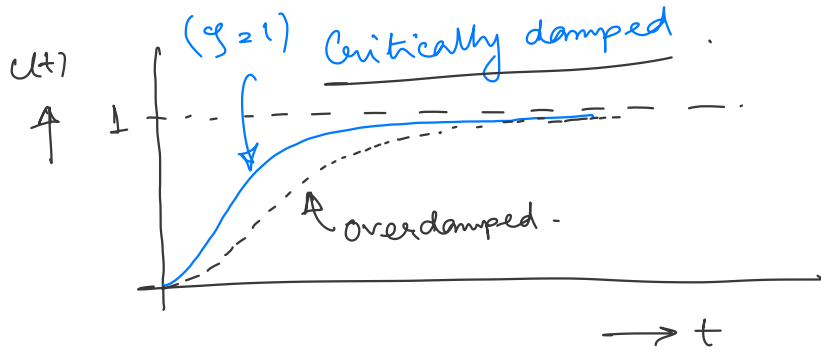
case - Ib Real repeated poles. ($\zeta = 1$)

$$\frac{\alpha}{2\sqrt{b}} = 1 \Rightarrow \zeta = 1$$

$$G(s) = \frac{\omega_n^2}{(s + \omega_n)^2}$$

$$C(s) = \frac{\omega_n^2}{s(s + \omega_n)^2} = \frac{1}{s} + \frac{k_1}{s + \omega_n} + \frac{k_2}{(s + \omega_n)^2}$$

$$\therefore c(t) = (1 + k_1 e^{-\omega_n t} + k_2 t e^{-\omega_n t}) u(t)$$



- Case-II : Complex conjugate poles:

$$0 < \frac{\alpha}{2\sqrt{b}} < 1 \equiv 0 < \zeta < 1$$

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}, \quad R(s) = \frac{1}{s}$$

$$\therefore C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} = \frac{K_1}{s} + \frac{K_2 s + K_3}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

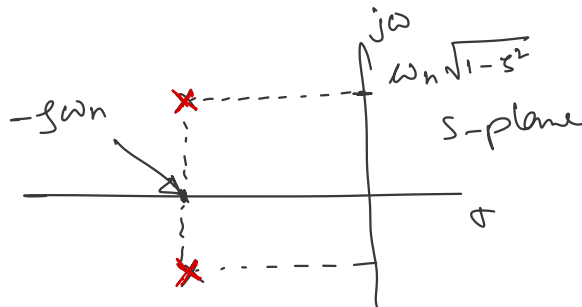
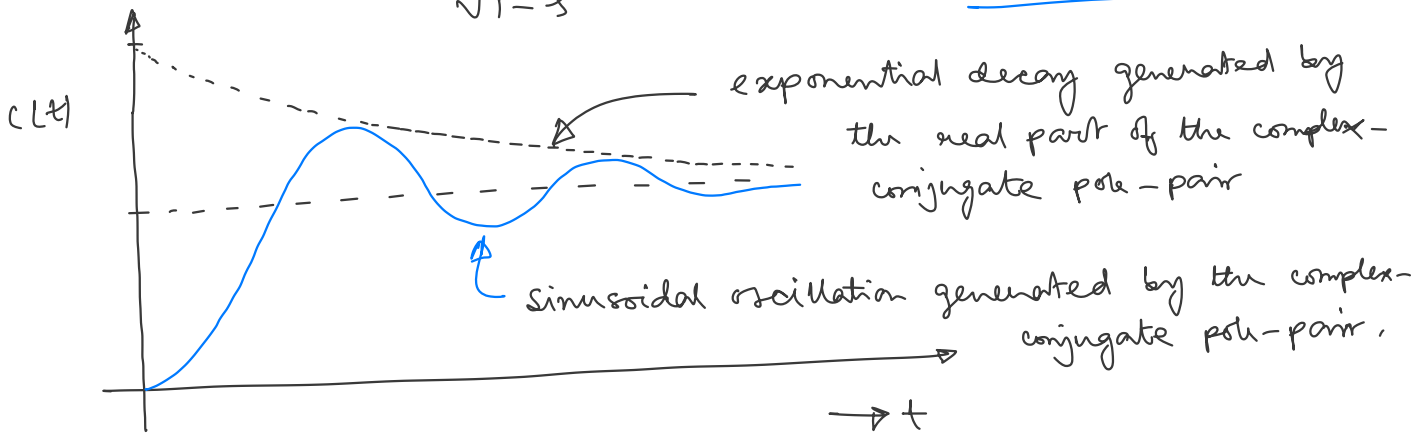
$$= \frac{1}{s} - \frac{(s + \zeta\omega_n) + \frac{\zeta}{\sqrt{1-\zeta^2}} \omega_n \sqrt{1-\zeta^2}}{(s + \zeta\omega_n)^2 + (\omega_n \sqrt{1-\zeta^2})^2}$$

$$\therefore c(t) = 1 - e^{-\zeta\omega_n t} \left[\cos \{ (\omega_n \sqrt{1-\zeta^2}) t \} + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \{ (\omega_n \sqrt{1-\zeta^2}) t \} \right]$$

$$c(t) = 1 - e^{-\zeta\omega_n t} \cdot \frac{1}{\sqrt{1-\zeta^2}} \cdot \cos \{ (\omega_n \sqrt{1-\zeta^2}) t - \phi \}, \quad \forall t \geq 0.$$

where, $\phi = \tan^{-1} \frac{\zeta}{\sqrt{1-\zeta^2}}$

"Underdamped"



o Case - III : Purely imaginary poles ($\zeta = 0$)

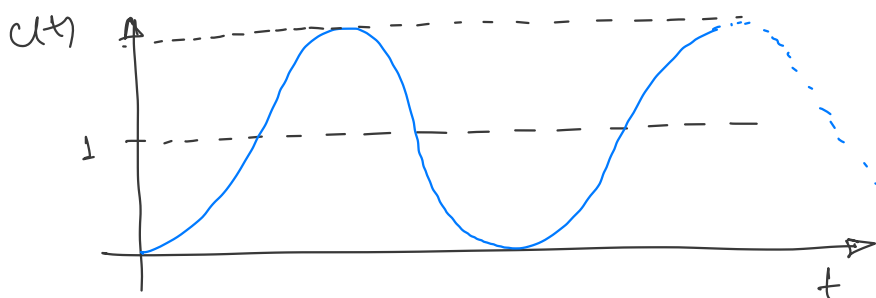
$$\alpha = 0 \equiv \zeta = 0$$

Poles: $\sigma_1, \sigma_2 = j\omega_n, -j\omega_n$

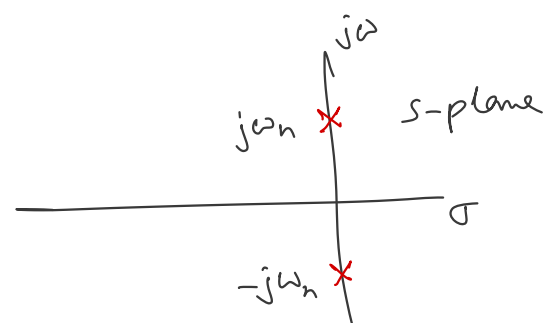
$$G(s) = \frac{\omega_n^2}{s^2 + \omega_n^2} ; \quad R(s) = \frac{1}{s}$$

$$C(s) = \frac{\omega_n^2}{s(s^2 + \omega_n^2)} = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}$$

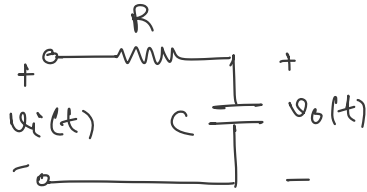
$$\therefore c(t) = [1 - \cos(\omega_n t)] u(t)$$



Undamped

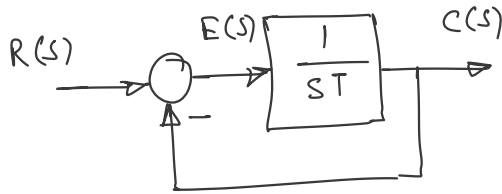


Time response (continued) - [Transient response]



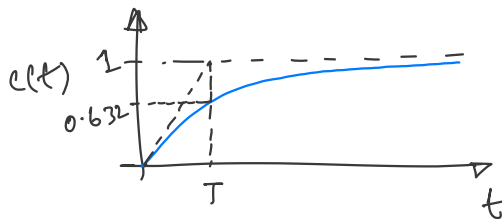
$$G(s) = \frac{1/RC}{s + 1/RC} ; \quad R(s) \rightarrow \boxed{G(s)} \rightarrow C(s)$$

Standard 1st order or 2nd order transfer functions may be encountered in closed-loop as well. $c(t) = ?$



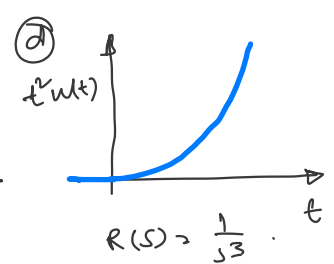
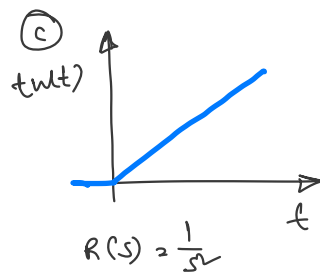
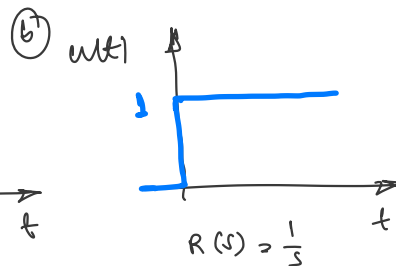
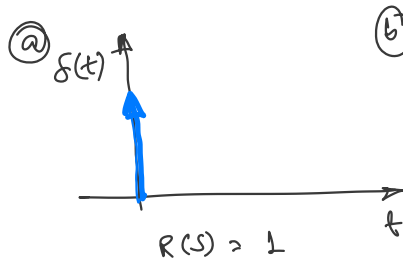
$$G(s) = \frac{1}{sT + 1} = \frac{1/T}{s + 1/T} = \frac{a}{s + a}$$

time constant : $\frac{1}{a}$ [e.g. RC , T]



$$c(t) = 1 - e^{-t/T}, \quad t \geq 0.$$

- Standard input signals for analysis of time response of LTI systems:



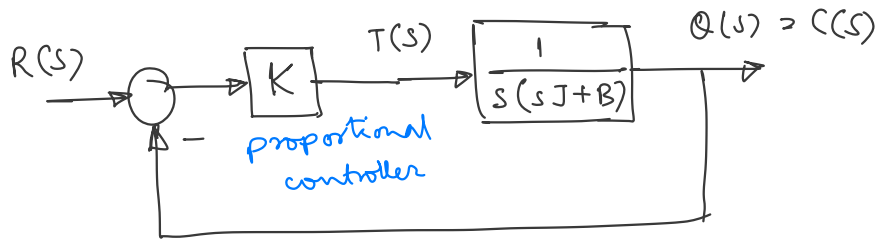
If the response due to one standard input signal is known for an LTI system, then we can easily obtain the response for other input signals by exploiting the linearity and time-invariance of the system.

⊗ Let a rotary motion system is described by:

$$J\ddot{\theta} + B\dot{\theta} = T$$

$$G(s) = \frac{Q(s)}{T(s)} = \frac{1}{s(sJ+B)}$$

A proportional controller is employed to track the angular position $\theta(t)$ to the reference $r(t)$.



$$\frac{C(s)}{R(s)} = \frac{k/J}{s^2 + (B/J)s + k/J}$$

$$\equiv \frac{\tilde{\omega}_n}{s^2 + 2\zeta\tilde{\omega}_n s + \tilde{\omega}_n^2}$$

("Standard form" of 2nd order system)

ω_n : undamped natural frequency

ζ : damping ratio: $\frac{\text{actual damping}}{\text{critical damping}} = \frac{\text{exponential decay freq.}}{\text{natural freq.}}$

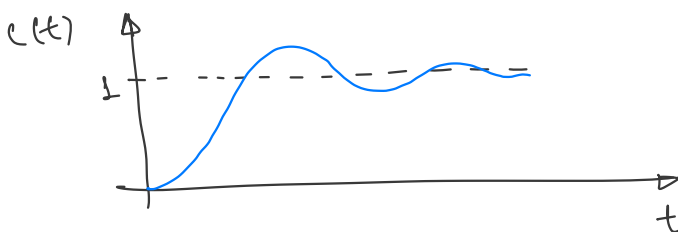
for $r(t) = u(t)$; we've seen

$$c(t) = 1 - e^{-\zeta\omega_n t} \left(\cos \omega_n \sqrt{1-\zeta^2} t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_n \sqrt{1-\zeta^2} t \right)$$

for $t \geq 0$.

$$= 1 - e^{-\zeta\omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right) \quad \text{for } t \geq 0.$$

where $\omega_d := \omega_n \sqrt{1-\zeta^2}$: damped natural frequency ($\zeta < 1$)



$$c(t) = 1 - e^{-\zeta\omega_n t} \cdot \frac{1}{\sqrt{1-\zeta^2}} \left(\sqrt{1-\zeta^2} \cos \omega_d t + \zeta \sin \omega_d t \right)$$

$$= 1 - e^{-\zeta\omega_n t} \cdot \frac{1}{\sqrt{1-\zeta^2}} \cos(\omega_d t - \phi)$$

where $\phi = \tan^{-1} \frac{\zeta}{\sqrt{1-\zeta^2}}$

$$= 1 - e^{-\zeta \omega_n t} \frac{1}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \psi),$$

where $\psi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$.

⊗ Transient response specifications:

- Rise time : (t_r) : time required for the step response to rise from 10% to 90% (overdamped case) or from 0% to 100% (underdamped case) of the final value.

$$c(t) = 1 - e^{-\zeta \omega_n t} \cdot \frac{1}{\sqrt{1-\zeta^2}} \left(\sqrt{1-\zeta^2} \cos \omega_d t + \zeta \sin \omega_d t \right)$$

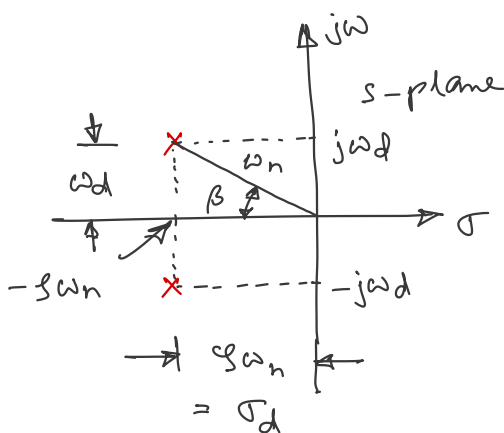
$$c(t=t_r) = 1 = 1 - e^{-\zeta \omega_n t_r} \cdot \frac{1}{\sqrt{1-\zeta^2}} \left(\sqrt{1-\zeta^2} \cos \omega_d t_r + \zeta \sin \omega_d t_r \right)$$

$$\sqrt{1-\zeta^2} \cos \omega_d t_r + \zeta \sin \omega_d t_r = 0$$

$$\Leftrightarrow \tan \omega_d t_r = - \frac{\sqrt{1-\zeta^2}}{\zeta} = - \frac{\omega_d}{\zeta \omega_n} = - \frac{\omega_d}{\sigma_d}$$

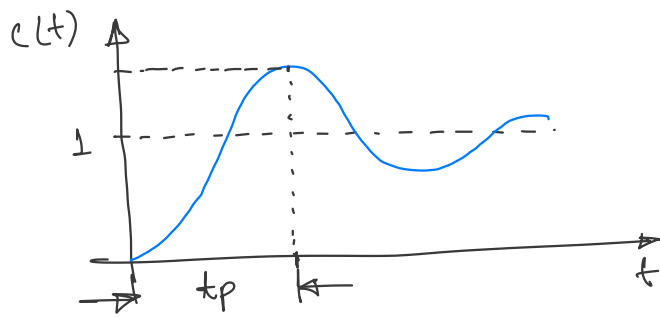
$$\left\{ \begin{array}{l} \omega_d := \omega_n \sqrt{1-\zeta^2} \\ \sigma_d := \zeta \omega_n \end{array} \right\}.$$

$$t_r = \frac{1}{\omega_d} \tan^{-1} \left(- \frac{\omega_d}{\sigma_d} \right)$$



$$\therefore t_r = \frac{1}{\omega_d} (\pi - \beta)$$

- Peak time : (t_p) : time required for the step response to reach the first peak of the overshoot.



$$c(t) = 1 - e^{-\zeta \omega_n t} \left(\cos \omega_d t + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t \right), \quad t \geq 0$$

$$\left. \frac{d(c(t))}{dt} \right|_{t=t_p} = 0$$

$$\begin{aligned} \therefore \left. \frac{d(c(t))}{dt} \right|_{t=t_p} &= \zeta \omega_n e^{-\zeta \omega_n t_p} \left(\cos \omega_d t_p + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t_p \right) \\ &\quad + e^{-\zeta \omega_n t_p} \left(\omega_d \sin \omega_d t_p - \frac{\zeta \omega_d}{\sqrt{1-\zeta^2}} \cos \omega_d t_p \right) \\ &= \sin \omega_d t_p \cdot \left(\frac{\zeta^2 \omega_n}{\sqrt{1-\zeta^2}} \cdot e^{-\zeta \omega_n t_p} - \frac{\zeta \omega_n \sqrt{1-\zeta^2}}{\sqrt{1-\zeta^2}} \cdot e^{-\zeta \omega_n t_p} \right) \end{aligned}$$

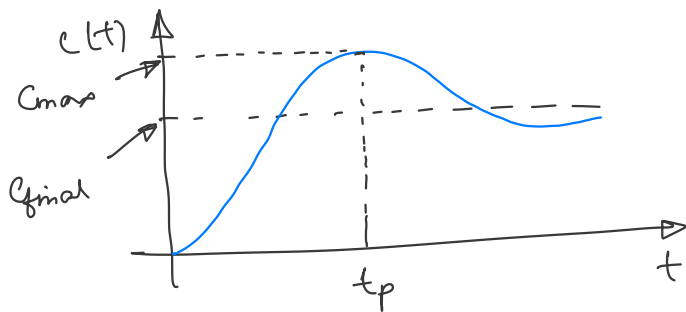
$$\therefore \left. \frac{d(c(t))}{dt} \right|_{t=t_p} = 0 \Rightarrow \sin \omega_d t_p = 0 \Rightarrow t_p = \frac{n\pi}{\omega_d}$$

$n = 1, 2, \dots$

for first peak, $n = 1 \Rightarrow \boxed{t_p = \frac{\pi}{\omega_d}}$

- % overshoot / Maximum percent overshoot : % OS / M_p

Amount that the step response overshoots the steady-state or final value at the peak time, expressed as a percentage of the steady-state value.



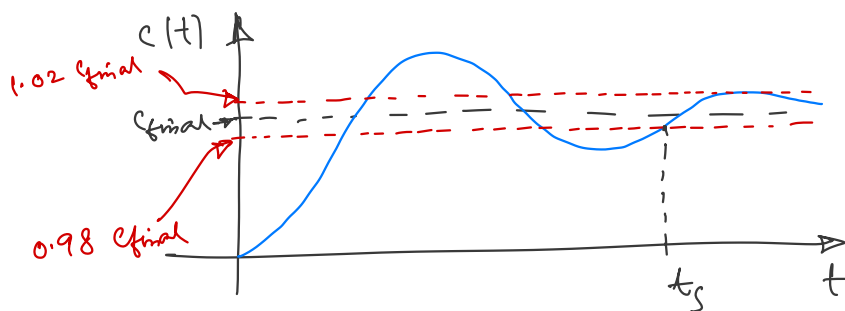
$$\%OS = \frac{c_{max} - c_{final}}{c_{final}} \times 100$$

$$\begin{aligned} c_{max} = c(t=t_p) &= 1 - e^{-\zeta \omega_n t_p} \left(\cos \omega_d t_p + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \omega_d t_p \right) \\ &= 1 - e^{-\zeta \omega_n \cdot \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}} \left(\cos \pi + \frac{\zeta}{\sqrt{1-\zeta^2}} \sin \pi \right) \quad (t \geq 0) \\ &= 1 + e^{\left(-\frac{\pi \zeta}{\sqrt{1-\zeta^2}} \right)} \end{aligned}$$

$$c_{final} = c(t \rightarrow \infty) = 1.$$

$$\therefore \%OS = e^{-\left(\pi \zeta / \sqrt{1-\zeta^2} \right)}$$

• Settling time : (t_s): time required by the step response to reach and stay within $\pm 2\%$ of the steady-state value.



$t_s \approx$ time for the decaying sinusoid to reach amplitude 0.02.

$$\therefore \frac{e^{-\zeta \omega_n t_s}}{\sqrt{1-\zeta^2}} = 0.02$$

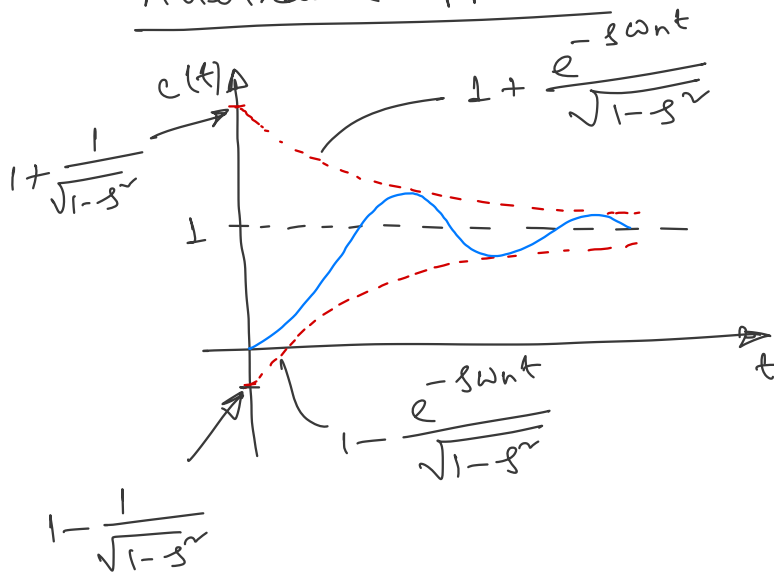
$$t_s = \frac{-\ln(0.02\sqrt{1-\zeta^2})}{\zeta\omega_n}$$

as ζ varies in $[0, 0.9]$

numerator of t_s is in $[3.91 \text{ to } 4.74]$

An approximate estimate : $t_s = \frac{4}{\zeta\omega_n}$

Alternative approach :



— time constant for the envelope functions = $\frac{1}{\zeta\omega_n} = T$

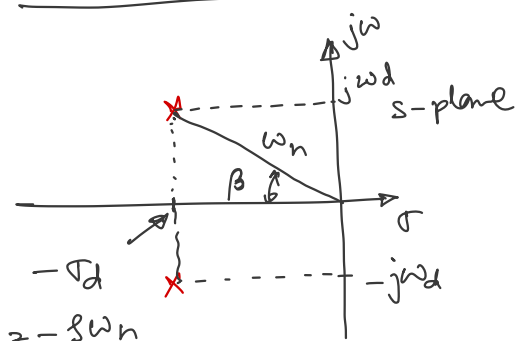
— speed of decay of $c(t)$ depends on the time-constant of these envelope curves

— at $4T$ it reaches within 98% of the final value.

$$t_s = \frac{4}{\zeta\omega_n} \quad (2\% \text{ band})$$

$$t_s = \frac{3}{\zeta\omega_n} \quad (5\% \text{ band})$$

(*) t_p , t_s , %OS in terms of pole locations.

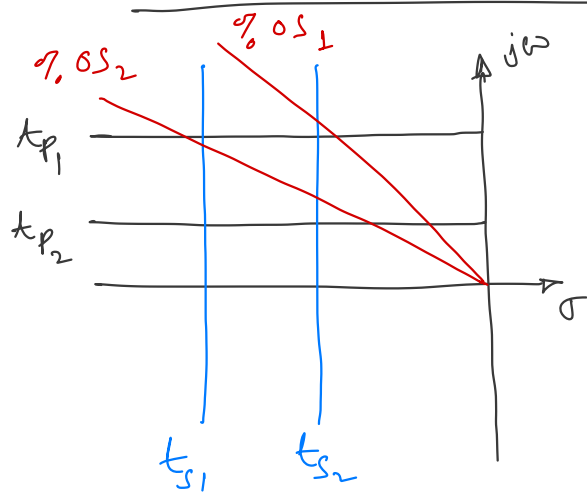


$$t_p = \frac{\pi}{\omega_d}$$

$$t_s = \frac{4}{\zeta\omega_n}$$

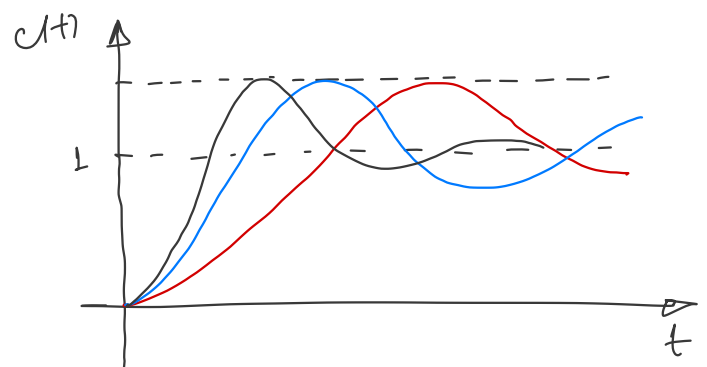
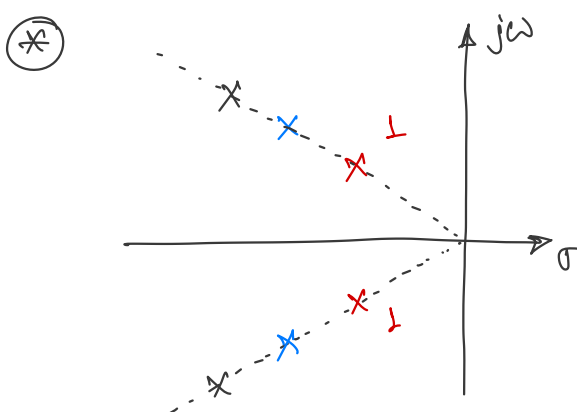
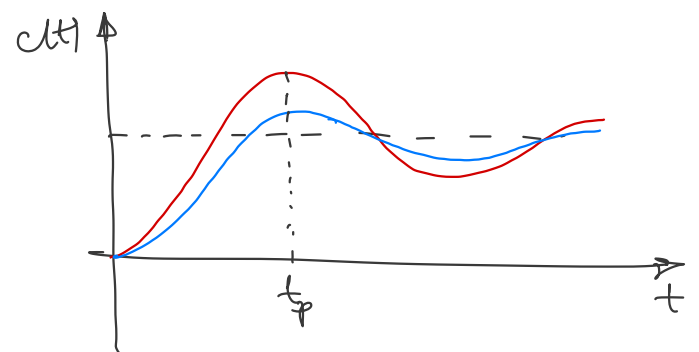
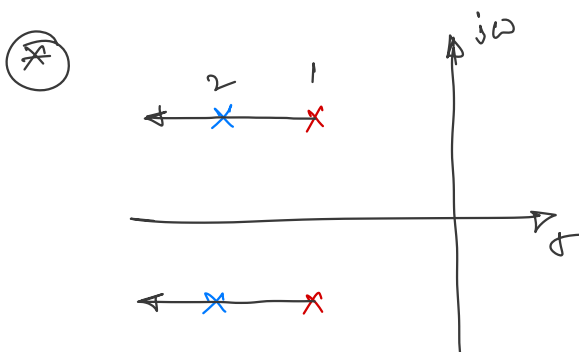
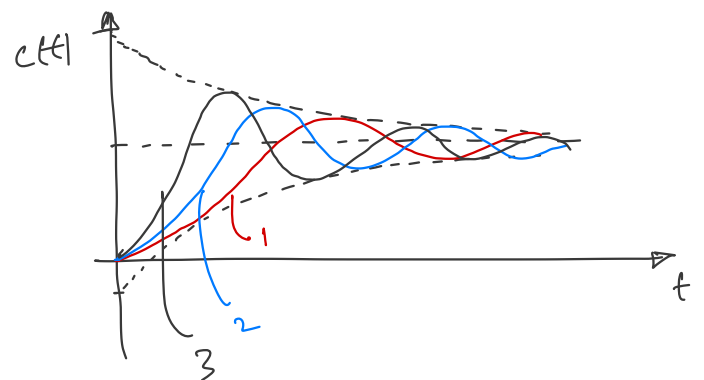
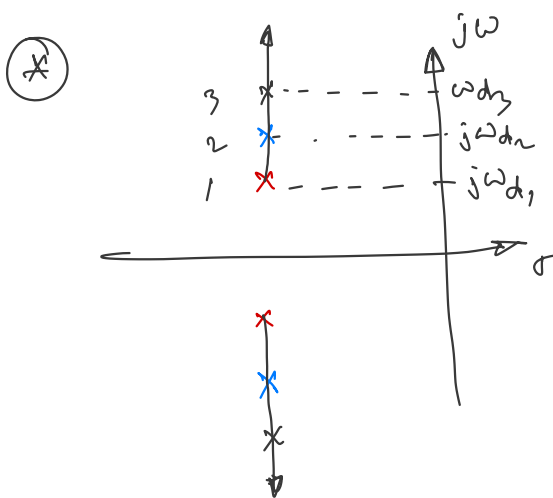
$$\%OS = e^{-\left(\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right)}$$

⊗ lines of constant t_p , t_s , and $\%OS$.



* $t_{s1} < t_{s2}$: $t_s \downarrow$ as poles move $\leftarrow$

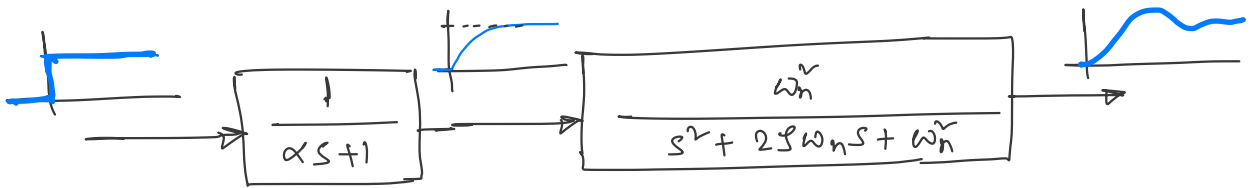
* $t_{p1} < t_{p2}$: $t_p \downarrow$ as $|\text{imag part}| \uparrow$



$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

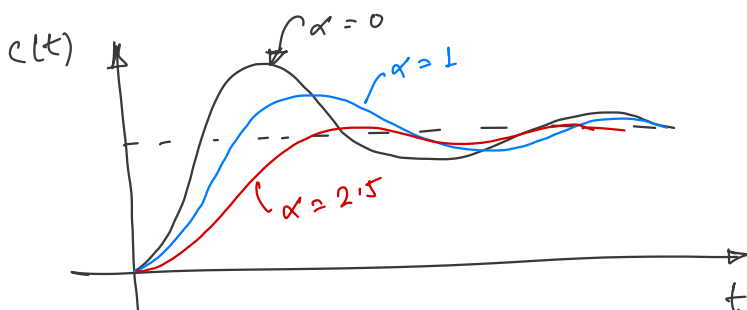
— Effect of additional pole (a real pole LHP)

$$\bar{G}(s) = \frac{\omega_n^2}{(\alpha s + 1)(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad , \text{ new pole } s = -\frac{1}{\alpha}$$



— slows down and smoothens the transients

- rise time increase
- overshoot decrease



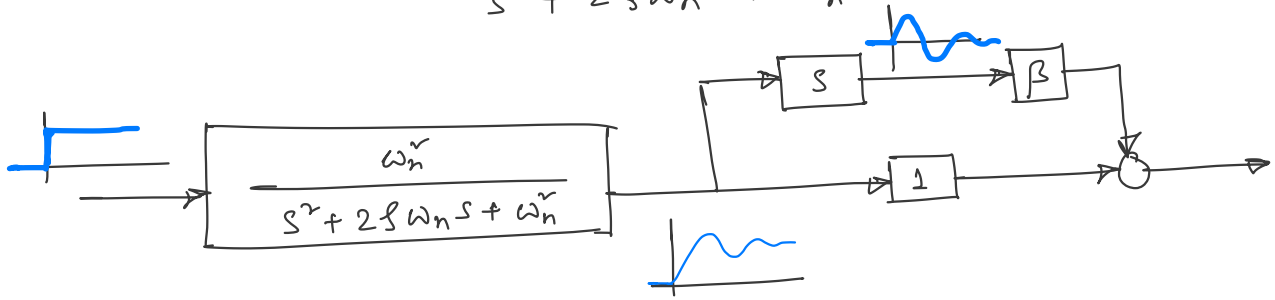
$0 < \alpha \ll 1$ — then the response approaches that of $\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$.

$$C(s) = \frac{A}{s} + \frac{B}{s + \frac{1}{\alpha}} + \frac{Cs + D}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

— If $-\frac{1}{\alpha} \ll -\zeta\omega_n$ — then the additional pole would not have much effect on $t_p, t_s, \%$ OS etc.

— Effect of additional zeros: (a real zero)

$$\bar{G}(s) = \frac{(\beta s + 1) \cdot \omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

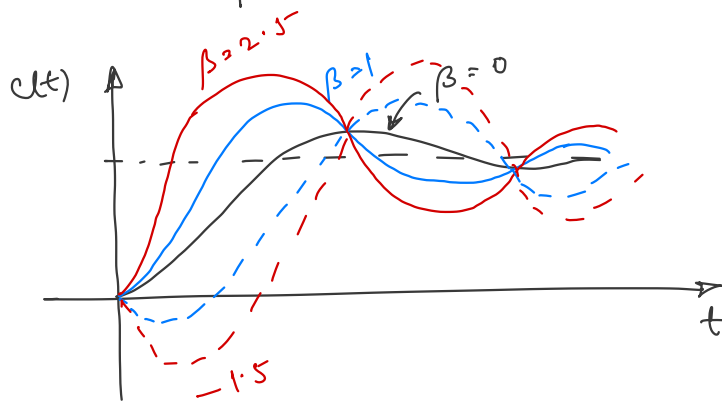


— decrease rise time

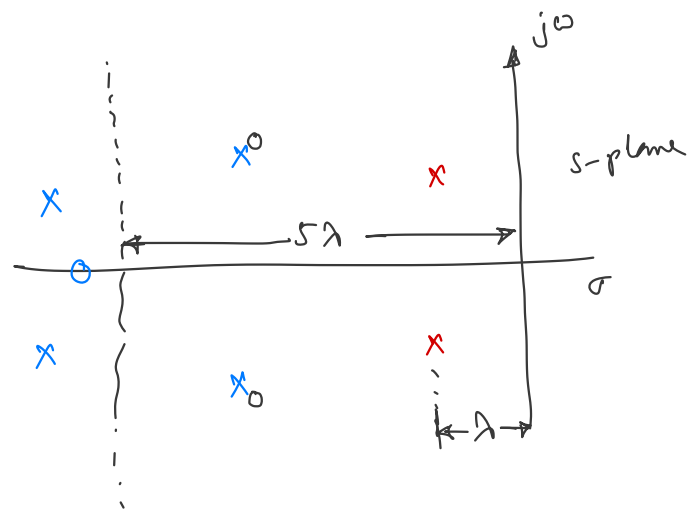
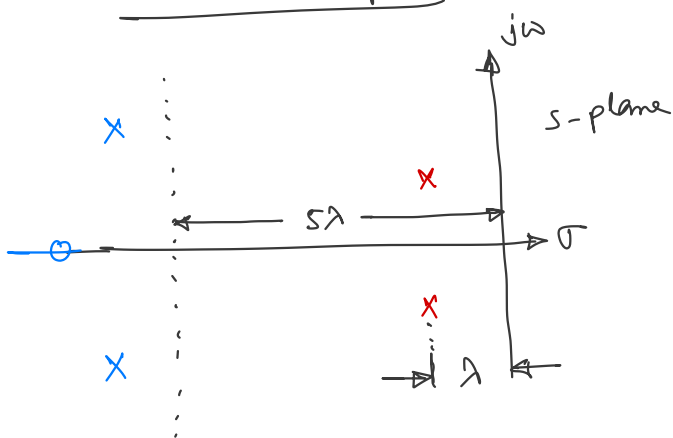
— increase the overshoot

If $|\beta| \ll 1$ — not much effect

— For $\beta < 0$, we will also have undershoot



— Dominant poles:



Rule of thumb: A group of poles/zeros is "dominant" if

- either the other poles are at least 5 times farther from the $j\omega$ -axis
- or the closer poles are "almost canceled" by zeros